

Session 8: FunctionsTopics:Declaration, Definition, Calling.Pass-by-value vs pass-by-reference.

Tasks

1.

Declare a function called `getUserInitials` that takes a user's full name (like 'Virat Kohli') and returns their initials in uppercase (e.g., 'VK'). Call this function with your favorite cricketer's name and print the result.

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
void getUserInitials(char name[], char initials[])
```

```
{
```

```
    int i = 0;
```

```
    initials[0] = toupper(name[0]);
```

```
    while(name[i] != '\0')
```

```
    {
```

```
        if(name[i] == ' ')
```

```
        {
```

```
            initials[1] = toupper(name[i + 1]);
```

```
            break;
```

```
        }
```

```
        i++;
```

```
    }
```

```
    initials[2] = '\0';
```

```
}
```

```

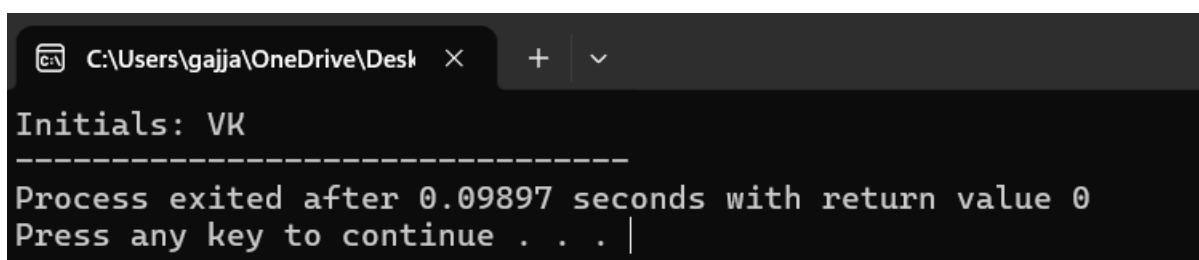
int main()
{
    char name[] = "Virat Kohli";
    char initials[3];

    getUserInitials(name, initials);

    printf("Initials: %s", initials);

    return 0;
}

```



```

C:\Users\gajja\OneDrive\Desktop
Initials: VK
-----
Process exited after 0.09897 seconds with return value 0
Press any key to continue . . . |

```

2.

Create a function `addToCart` that takes a shopping cart array and a product name, adds the product to the cart, and prints the updated cart. Demonstrate how passing the cart array by reference allows changes to persist outside the function.

Hint: In languages like JavaScript, arrays are passed by reference. In C/C++, use pointers for reference behavior.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
void addToCart(char cart[][50], int *count, char product[])
```

```

{
    strcpy(cart[*count], product);
    (*count)++;
}

```

```
printf("\nUpdated Cart:\n");
```

```
for(int i = 0; i < *count; i++)
```

```
{
```

```
    printf("%d. %s\n", i + 1, cart[i]);
```

```
}
```

```
}
```

```
int main()
```

```
{
```

```
    char cart[10][50];
```

```
    int count = 0;
```

```
    addToCart(cart, &count, "Laptop");
```

```
    addToCart(cart, &count, "Mouse");
```

```
    addToCart(cart, &count, "Keyboard");
```

```
    return 0;
```

```
}
```

```
C:\Users\gajja\OneDrive\Desk  ×  +  v

Updated Cart:
1. Laptop

Updated Cart:
1. Laptop
2. Mouse

Updated Cart:
1. Laptop
2. Mouse
3. Keyboard

-----
Process exited after 0.1229 seconds with return value 0
Press any key to continue . . . |
```

3.

Write two functions: `increaseFollowersByValue` and `increaseFollowersByReference`. Each should take a followers count and add 1000 to it, but one should use pass-by-value and the other pass-by-reference. Show how the original followers count changes (or doesn't) after calling each function.

```
#include <stdio.h>
```

```
void increaseFollowersByValue(int followers)
```

```
{
```

```
    followers = followers + 1000;
```

```
    printf("Inside function: %d\n", followers);
```

```
}
```

```
int main()
```

```
{
```

```
    int followers = 5000;
```

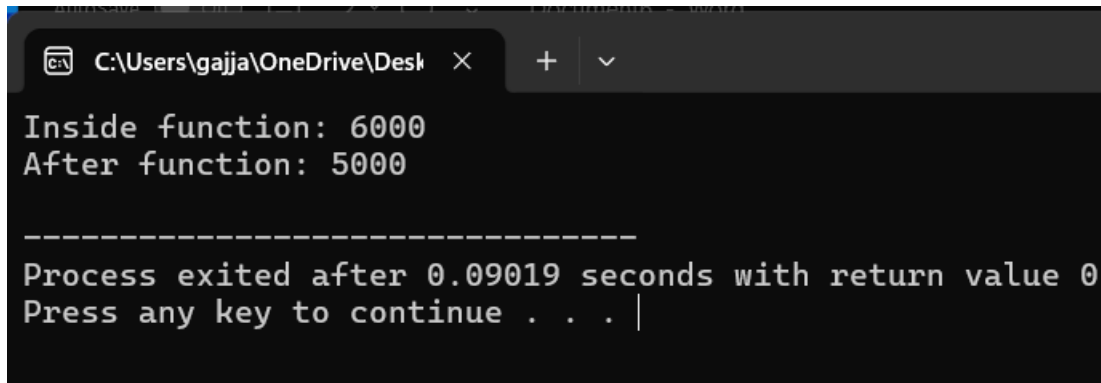
```

    increaseFollowersByValue(followers);

    printf("After function: %d\n", followers);

    return 0;
}

```



```

C:\Users\gajja\OneDrive\Desktop
Inside function: 6000
After function: 5000

-----
Process exited after 0.09019 seconds with return value 0
Press any key to continue . . . |

```

4.

Build a function `formatPrice` that takes a price in rupees and returns a string formatted like Flipkart's price tag (e.g., '₹1,599'). Use this function to display prices for three different products.

```
#include <stdio.h>
```

```

void formatPrice(int price)
{
    if(price >= 100000)
    {
        printf("₹%d,%02d,%03d", price / 100000,
            (price / 1000) % 100,
            price % 1000);
    }
    else if(price >= 1000)
    {

```

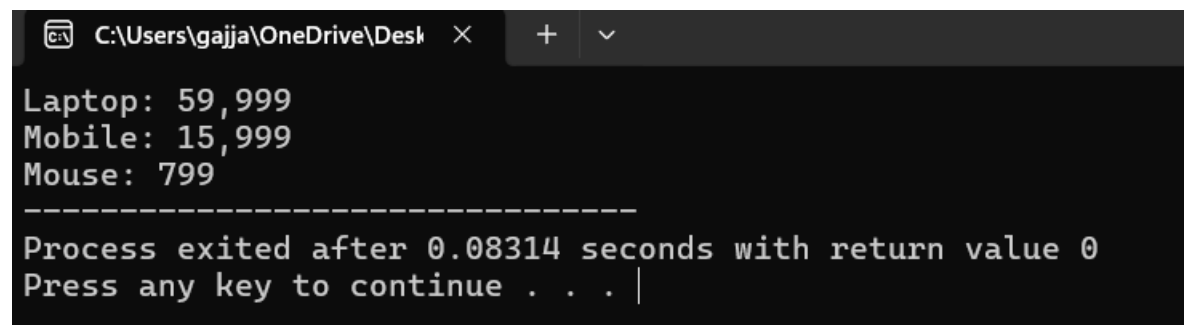
```
        printf("₹%d,%03d", price / 1000, price % 1000);
    }
    else
    {
        printf("₹%d", price);
    }
}
```

```
int main()
{
    printf("Laptop: ");
    formatPrice(59999);

    printf("\nMobile: ");
    formatPrice(15999);

    printf("\nMouse: ");
    formatPrice(799);

    return 0;
}
```



```
C:\Users\gajja\OneDrive\Desktop  X  +  v
Laptop: 59,999
Mobile: 15,999
Mouse: 799
-----
Process exited after 0.08314 seconds with return value 0
Press any key to continue . . . |
```

5.

Refactor an existing function you wrote above to make it reusable for both product names and usernames (for example, a function that capitalizes the first letter of any string).

Constraint: The refactored function should work for any string input, not just a specific use case.

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
void capitalizeFirstLetter(char text[])
```

```
{  
    if(text[0] >= 'a' && text[0] <= 'z')  
    {  
        text[0] = toupper(text[0]);  
    }  
}
```

```
int main()
```

```
{  
    char product[] = "laptop";  
    char username[] = "manan";  
  
    capitalizeFirstLetter(product);  
    capitalizeFirstLetter(username);  
  
    printf("Product: %s\n", product);  
    printf("Username: %s", username);  
  
    return 0;  
}
```



C:\Users\gajja\OneDrive\Desk



Product: Laptop

Username: Manan

Process exited after 0.09848 seconds with return value 0

Press any key to continue . . . |